

```
cd kinect
git clone git://github.com/OpenNI/OpenNI.git
cd OpenNI
git checkout unstable
```



Note: The version mentioned in the README file should at least be 1.5.4.0.

The software is set up for software floating point, but Ubuntu has been compiled for hardware floating point—so the compiler flags need to be modified. To do this, issue the following commands:

```
cd Platform/Linux/Build
sudo gedit Common/Platform.Arm
```

Remove the “*-mfloat-abi*” option in the file, so that it finally looks like what’s shown below:

```
ifeq "$(CFG)" "Release"
# Hardware specifying flags
CFLAGS += -march=armv7-a -mtune=cortex-a8 -mcpu=neon
#-mcpu=cortex-a8
# Optimisation level, minus currently buggy optimising methods
(which break bit-exact)
CFLAGS += -O3 -fno-tree-pre -fno-strict-aliasing
# More optimisation flags
CFLAGS += -ftree-vectorize -ffast-math -funsafe-math-
optimizations -fsingle-precision-constant
endif
```

To build OpenNI, run the following:

```
cd ~/kinect/OpenNI/Platform/Linux/CreateRedist
./RedistMaker.Arm
```



Note: It’s possible that this will result in an error because *MAKE_ARGS* includes “*-j0*”, which is illegal. If this occurs, edit *Redist_OpenNI.py* and find the line that looks like the one below:

```
MAKE_ARGS += ' -j ' + calc_jobs_number()
```

...and change it to...

```
MAKE_ARGS += ' -j1'
```

This will build the OpenNI binaries and create a folder called *Redist*. Then install OpenNI with the following command:

```
cd ~/kinect/OpenNI/Platform/Linux/Redist/OpenNI-Bin-Dev-Linux-
Arm-v1.5.4.0
sudo ./install.sh
```



Note: The version number in the path may need to be changed to reflect whatever is the current version.

For the second part (installing the Kinect driver), first download the source code, as follows:

```
cd ~/kinect
git clone git://github.com/avin2/SensorKinect.git
```

Next, configure the compiler flags; the software floating point option has to be turned off. In the *~/kinect/SensorKinect/Platform/Linux/Build/Common* directory, edit the *Platform.Arm* file to remove the “*-mfloat-abi*” option, as we did earlier for OpenNI.

Now, build the driver, which will create a *Redist* folder:

```
cd ~/kinect/SensorKinect/Platform/Linux/CreateRedist
./RedistMaker
```

To configure the USB port, an edit is required to the config file, or else it will not select the correct USB port. Edit *Config/GlobalDefaultsKinect.ini* in *~/kinect/SensorKinect/Platform/Linux/Redist/Sensor-Bin-Linux-Arm-v5.1.2.1* and find the line that reads:

```
;UsbInterface=2
```

Change this line (un-comment and change the interface from 2 to 1) to:

```
UsbInterface=1
```

Finally, install the driver and test it:

```
cd ~/kinect/SensorKinect/Platform/Linux/Redist/Sensor-Bin-Linux-
Arm-v5.1.2.1
sudo ./install.sh
```

```
cd ~/kinect/OpenNI/Platform/Linux/Redist/OpenNI-Bin-Dev-Linux-
Arm-v1.5.4.0
cd Samples/Bin/Arm-Release
./Sample-NiSimpleRead
```

This should result in a series of values being displayed, which vary if something is waved in front of Kinect.

Install OpenCV, with Kinect support, on the same platform



Note: These instructions would also work for Ubuntu running on a desktop machine or a laptop. For PandaBoard, it took me about 2.5 hours, whereas on my i7 machine, it took 15 minutes!

Approach 1: If you are not interested in installing the latest